

Systematic Review (PRISMA)

Dissertation (EI or ETI)

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What is ...?

“A review that is conducted according to clearly stated, scientific research methods, and is designed to minimize biases and errors inherent to traditional, narrative reviews.”

(source: Margalio, Zvi, Kevin C. Chung. Systematic Reviews: A Primer for Plastic Surgery Research. PRS Journal. 120/7 (2007))

What is...?

- Systematic reviews ask a specific question about the effectiveness of a treatment and answer it by summarising evidence that meets a set of pre-specified criteria.
- The process starts with a research question and a protocol or research plan . A review team searches for studies to answer the question using a highly sensitive search strategy. The retrieved studies are then screened for eligibility using the inclusion and exclusion criteria (this is done by at least two people working independently  ). Next, the reviewers extract the relevant data and assess the quality of the included studies. Finally, the review team synthesises the extracted study data and presents the results.

(source https://www.covidence.org/blog/the-difference-between-a-systematic-review-and-a-scoping-review/?campaignid=13271466382&adgroupid=123024098859&adid=524233276729&gclid=Cj0KCQjwkt6aBhDKARIsAAyeLJ14bSKsh6tminw9KbW4ofPf_3m_uHPRvQOVcggguFci-xyXvzhpAfClaAhjcEALw_wcB)

What is ...?

IDENTIFY THE RESEARCH QUESTION

DEFINE INCLUSION & EXCLUSION CRITERIA

SEARCH FOR STUDIES

SELECT STUDIES

EXTRACT DATA

ASSESS QUALITY

SYNTHESISE & PRESENT RESULTS
(E.G. USING META-ANALYSIS)

Systematic ...

- means that the methods used to search for and analyse the data are:
 - transparent, reproducible and defined before searching begins.
 - Systematic reviews strive to be as thorough and rigorous as possible to minimise the bias that would result from cherry-picking studies in a non-systematic way.

Purpose of a Systematic Review

- Summarizes and synthesizes existing knowledge
- Enables researchers to make an objective assessment of primary research evidence in order to inform or support further research

Key Characteristics

- Clearly identify title and goals
- Defined strategy to apply search
- Explicit and grounded criteria for inclusion or exclusion of any study
- Clear presentation of characteristics of each study
- Analysis of methodological quality
- Complete list of excluded studies and justifications

Key Characteristics

- Clear analysis of the included studies
 - Quantitative or qualitative analysis if appropriated
- Production of a report
 - Identifying the aim
 - Describing the methods and materials
 - Reporting results

When starting what to do ...

- Formulate a questions
- Perform the search
- Refine the results applying the inclusion and exclusion criteria
- Gather data and evaluate quality and validity
- Synthesizes, interprets and report data

Hypothesis

- “A systematic review should be based on principles of hypothesis testing, and the hypotheses must be conceived a priori.”

- (source: Margalioth, Zvi, Kevin C. Chung. Systematic Reviews: A Primer for Plastic Surgery Research. PRS Journal. 120/7 (2007))

Hypothesis



- The hypothesis is a prediction, but it involves more than a guess. Most of the time, the hypothesis begins with a question which is then explored through background research. It is only at this point that researchers begin to develop a testable hypothesis.
- Unless you are creating an exploratory study, your hypothesis should always explain what you expect to happen.

Focus of the Question

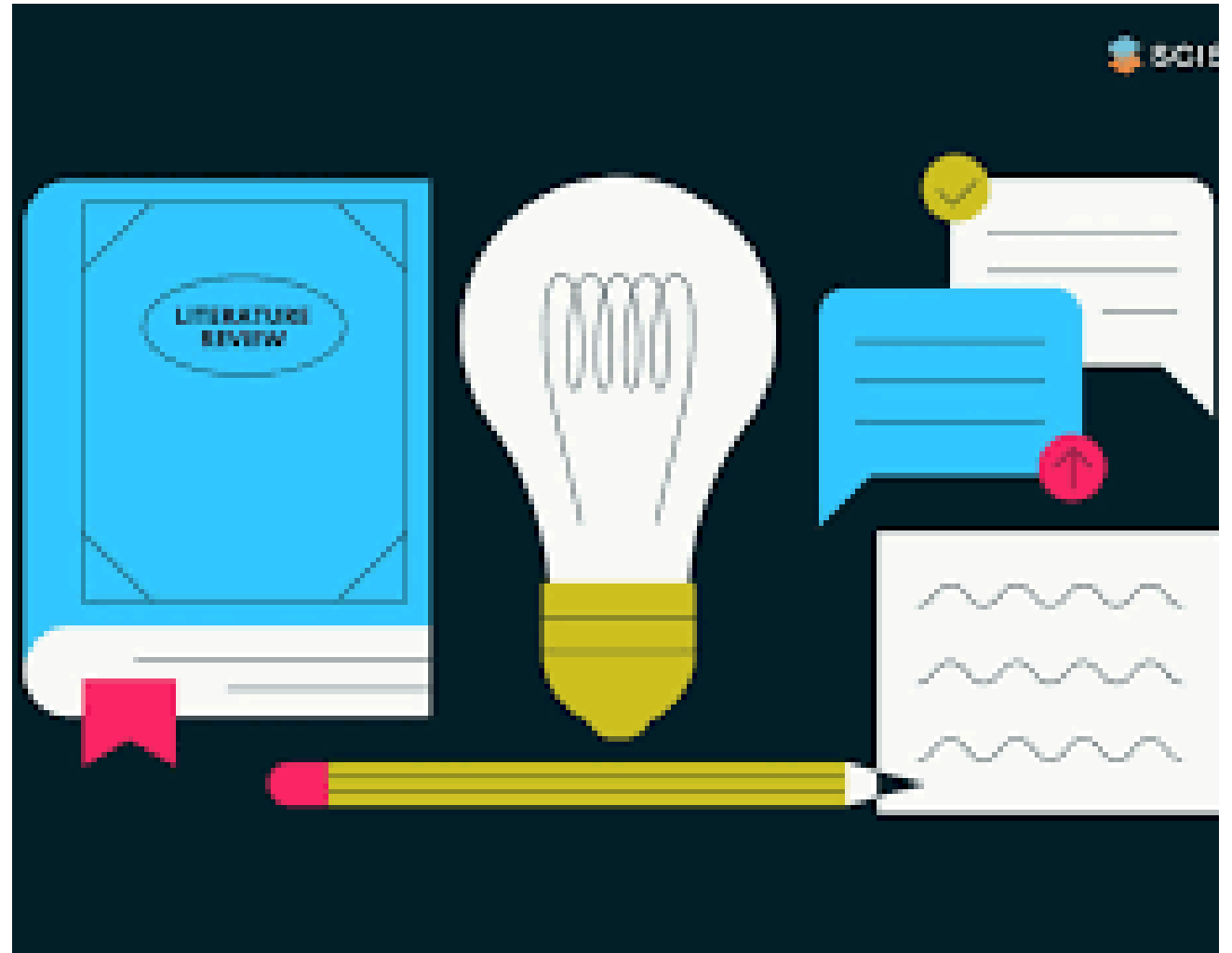
- The structured question will determine the inclusion and exclusion criteria:
 - What is the population of interest?
 - What are the interventions?
 - What are the outcomes of interest?
 - What study designs are appropriate?

Inclusion/Exclusion Criteria

- After establishing the purpose and hypothesis of the study, a comprehensive list of inclusion and exclusion criteria should be produced
- To avoid bias establish the list before starting the search

Literature Search

The basis of a systematic literature review is a comprehensive and repeatable literature search



Pitfalls

Database Bias – do not limit to one single database source

Publication Bias – publish several sources: journals; conferences; workshops and reports

English- language Bias not considering papers which are not written in English

Citation Bias – occurs when studies with significant or positive results are referenced only in papers that are not relevant

Data Collection

- The characteristics of the data collection should be established at the design stage of the study.
- Data should include:
 - Study characteristics
 - Sample Demographics (if applied!)
 - Outcome data
- There should be a protocol to avoid missing data
- Ideally it should be performed by two researchers

Quality Assessment

- The validity of a systematic review depends on the scientific method applied in the studies and on the reporting of data.

Types of Systematic Reviews

- By Joanna Briggs Institute (<https://jbi.global/>)
 - 1. Effectiveness
 - 2. Qualitative Research
 - 3. Economic Evaluations
 - 4. Prevalence/ Incidence
 - 5. Etiology/Risk
 - 6. Mixed Methods
 - 7. Umbrellas/ Overviews (reviews of reviews)
 - 8. Text/ Opinion
 - 9. Diagnostic Test Accuracy
 - 10. Scoping (mapping) Reviews

Limitations

- They encounter the same pitfalls as primary research.
 - Repeatability
 - Bias (publication, etc.)
 - Appropriate uses of methodology and statistics
- Small effects of interventions/research may be magnified.
- SRs are difficult to find (or execute) when evidence is scarce.
- Many SRs do not show the protocol used.
 - Questionable validity, reproducibility
- Considered as being a wasteful use of time and resources if poorly done

PRISMA

- An international, collaborative effort to address subpar reporting of review reports.
- 1996: QUORUM (Quality of Reporting Meta-analyses of RCTs)
- 2009: PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) is a result of QUORUM revisions and additions.
- Intended to be a living document

Goals

- Improve reporting of SRs and meta-analyses
- Useful for critically appraising other SRs (but is not to be used as a quality assessment instrument).

(Moher, Liberati, Tetzlaff, Altman, & The PRISMA Group, 2009)

Elements

PRISMA Statement

27 item checklist

4 phase flow diagram

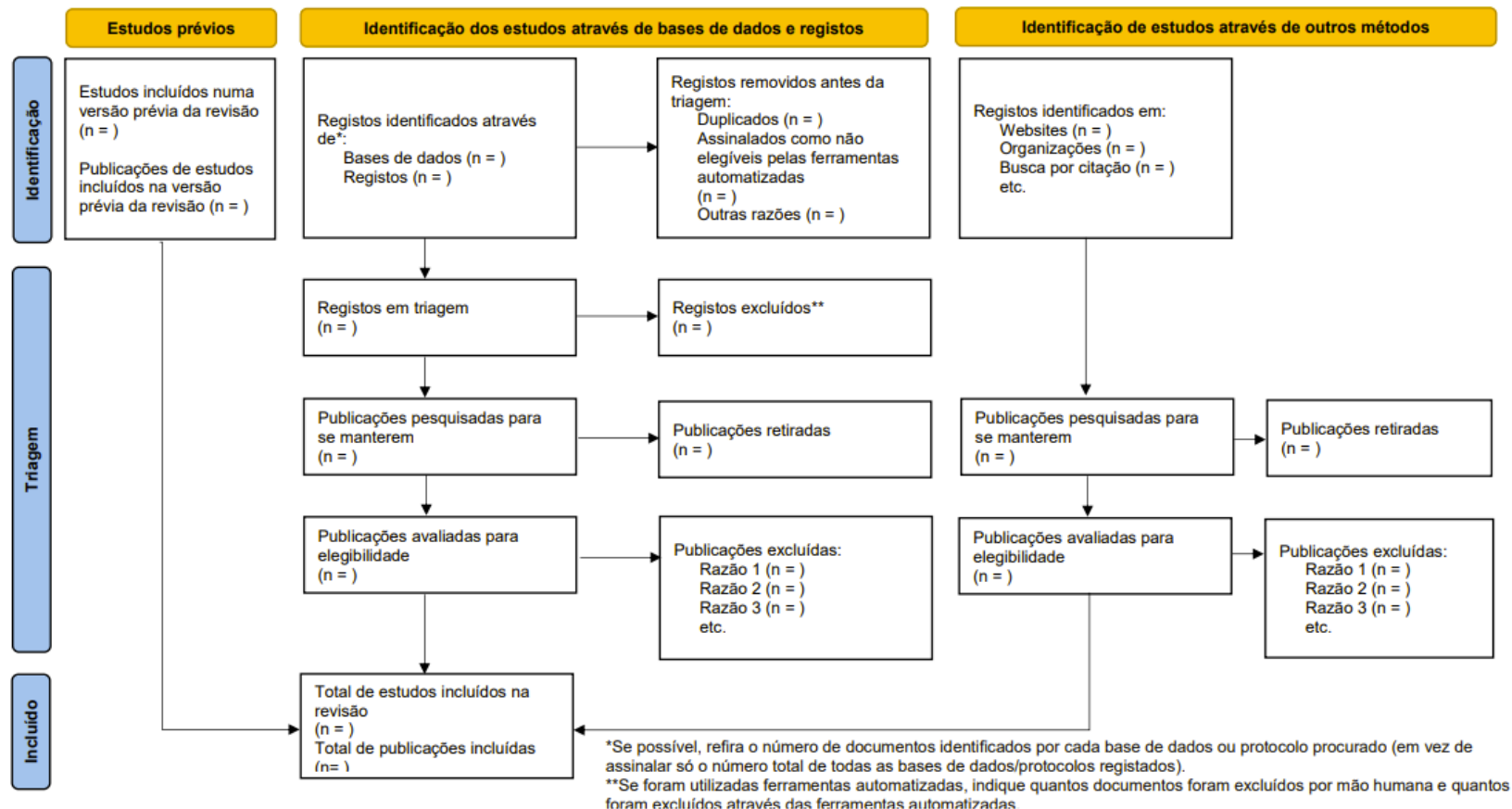


PRISMA 2020 Lista de verificação

Secção e Tópico	Item #	Verificação do item	Local onde o item está
TÍTULO			
Título	1	Identifica a publicação como uma revisão sistemática.	
RESUMO			
Resumo	2	Ver a lista de verificação PRISMA 2020 para Resumos.	
INTRODUÇÃO			
Fundamentação	3	Fundamenta a revisão no contexto do conhecimento existente.	
Objetivos	4	Apresenta explicitamente o(s) objetivo(s) ou questão(ões) respeitantes à revisão.	
MÉTODOS			
CrITÉRIOS de elegibilidade	5	Especifica os critérios de inclusão e exclusão para a revisão e forma como os estudos foram agrupados para as sínteses.	
Fontes de informação	6	Especifica todas as bases de dados, registos, websites, organizações, listas de referências e outras fontes pesquisadas ou consultadas para identificação dos estudos. Especifica a última data em que cada fonte foi pesquisada ou consultada.	
Estratégia de pesquisa	7	Apresenta as estratégias de pesquisa completas para todas as bases de dados, registos e websites, incluindo todos os filtros e limites utilizados.	
Processo de seleção	8	Especifica os métodos utilizados para decidir se um estudo satisfaz os critérios de inclusão da revisão, incluindo quantos revisores fizeram a triagem de cada registo e publicação selecionada, se trabalharam de uma forma independente e, se aplicável, os detalhes de ferramentas de automatização utilizadas no processo.	
Processo de recolha de dados	9	Especifica os métodos utilizados para recolha de dados das publicações, incluindo quantos revisores recolheram a informação de cada publicação, se trabalharam de uma forma independente, todos os processos de obtenção ou confirmação de dados por parte dos investigadores do estudo e, se aplicável, detalhes de ferramentas de automatização utilizadas.	
Dados dos itens	10a	Lista e define todos os resultados para os quais os dados foram pesquisados. Especifica se foram pesquisados todos os resultados compatíveis com cada domínio em cada estudo (p ex. para todas as medidas, momentos, análises) e, se não, especifica os métodos utilizados para decidir quais resultados a recolher.	
	10b	Lista e define todas as outras variáveis para as quais os dados foram pesquisados (p. ex. características dos participantes e intervenções, fontes de financiamento). Descreve os pressupostos utilizados sobre informação em falta ou pouco clara.	
Avaliação do risco de viés nos estudos	11	Especifica os métodos utilizados para avaliar o risco de viés dos estudos incluídos, incluindo detalhes sobre o(s) instrumento(s) utilizado(s), quantos revisores avaliaram cada estudo e se trabalharam de forma independente e ainda, se aplicável, detalhes de ferramentas de automatização utilizadas no processo.	

Flow

PRISMA 2020 Fluxograma para novas revisões sistemáticas que incluam buscas em bases de dados, protocolos e outras fontes



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de: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

Para mais informações, visite: <http://www.prisma-statement.org/>

References and Links:

- <https://guides.lib.unc.edu/systematic-reviews/overview>
- <https://www.prisma-statement.org/documents/PRISMA%202020%20flow%20diagram%20EUROPEAN%20PORTUGUESE.pdf>
- https://www.slideshare.net/kpadron_libraries/prisma-and-systematic-reviews-135954954